CELL: THE UNIT OF LIFE

CELL

Cellular Organisation is a defining feature of living organisms.

- Organisms
  - Unicellular → Single celled → Ex. Bacteria, Amoeba
  - Multicellular → Many celled → Ex. Humans

Unicellular → capable of independent existance.
→ capable of performing all essential functions of life.

* Anything less than a complete cell can't live independently
* Cell is fundamental unit of life.
  - Structural unit and Functional unit

- Robert Hooke : Discovered cell (dead cell)
- Antonievon Leeuwenhoek : Discovered live cell
- Robert Brown (1831) : Discovered nucleus
- electron microscope (1950) : Details of cells like cell membrane & ribosomes were seen.

CELL THEORY

Matthias Schleiden (1838)	Theodore Schwann (1839)
German botanist	German zoologist
Examined many plants & observed that all plants are made up of different cells which make different tissues.	• Animal cells → surrounded by thin outer layer (Today known as plasma/cell membrane) • Plant cells → surrounded by CW ☆ Hypothesis : All plants & animals are made of cells and products of cells.

tissues. of cells. membrane

Schleiden + Schwann

Cell theory

But it didn't explain how new cells are formed.

Rudolf Virchow (1855) : Firstly explained that cells divide and new cells are formed from pre-existing cells (Omnis cellula e cellula)

Modified hypothesis given by Schleiden & Schwann and give final shape to cell theory :→
→ all living organisms are composed of cells and products of cells.
→ all cells arise from pre-existing cells.

AN OVERVIEW OF CELL

Plant cell

Ex. Onion peel cell

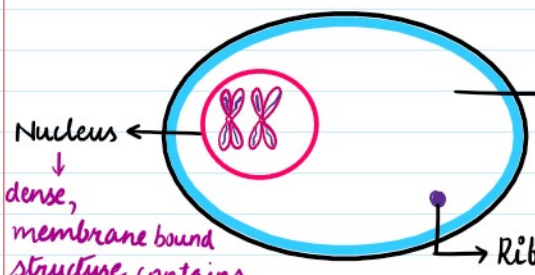
→ Outermost layer is cell wall
→ Cell wall ⊕nt.
→ Chloroplast ⊕nt.
→ central vacuole ⊕nt.
→ centriole ⊖nt

May occupy upto 90% volume

Animal cell

Ex. Human cheek cell

→ Outermost layer is cell membrane
→ cell wall ⊖nt.
→ chloroplast ⊖nt.
→ central vacuole ⊖nt.
→ centriole ⊕nt



Nucleus → dense, membrane bound structure contains chromosomes that infact contains DNA.

Cytoplasm : semi-fluid matrix occupy volume of cell ; main arena of cellular activities ; keep cell in living state ; universal

Ribosome : Non-membrane bound particles ; Universal
- Can be found in cytoplasm, associated to RER, in mitochondria & chloroplast.
- In prokaryotes, associated with cell memb.
- called : Palade particles (George Palade 1953)
  : Protein factory (mRNA to protein)

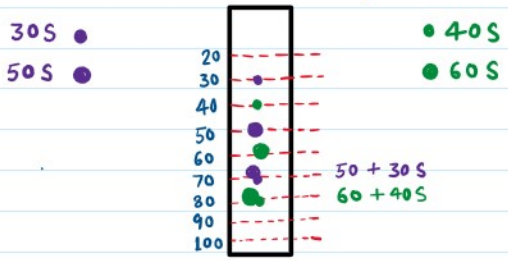
- In prokaryotes, associated with cell membrane.
- called: Palade particles (George Palade 1953)
  : Protein factory (mRNA to protein)

Polysome — Ribosome, mRNA

: Smallest organelle (15-20 nm)

- Ribosome = mRNA + Protein
- 70S (L-50S, S-30S) – Prokaryotes
  – mitochondria, – chloroplast } Eukaryotes
- 80S (L-60S, S-40S) – Eukaryotic cytoplasm

↳ Svedberg unit / sedimentation coefficient
- Indirect measure of density and size.

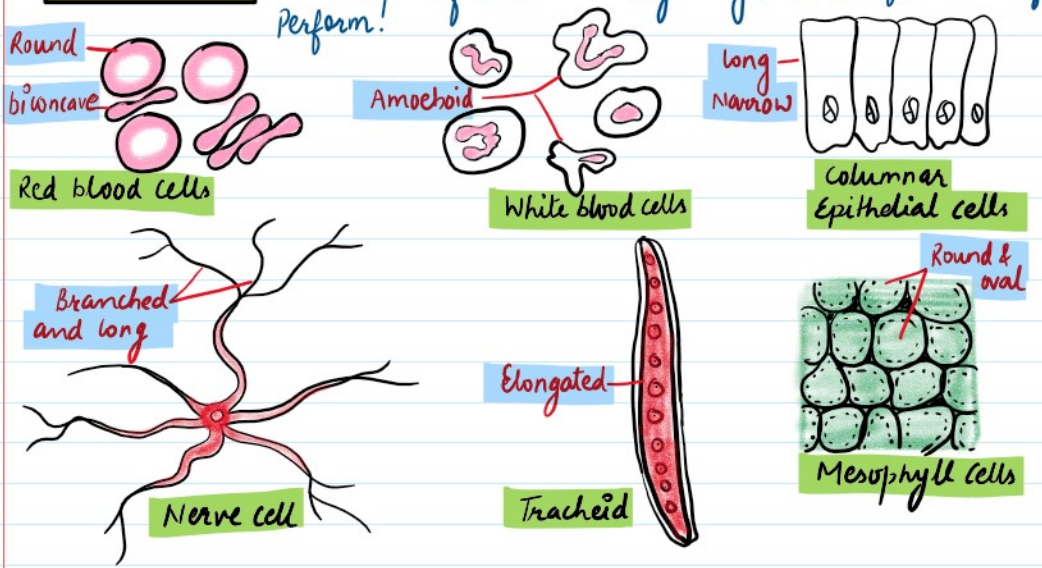


CELL SIZE

Cell	Feature
Mycoplasma	Length = 0.3 μm
PPLO	Size = 0.1 μm
Virus	Size = 0.02 - 0.2 μm
Ostrich Egg	Largest cell
Typical Eukaryotic cell	10 - 20 μm

CELL SHAPE

The shape of the cell may vary with the function they perform!

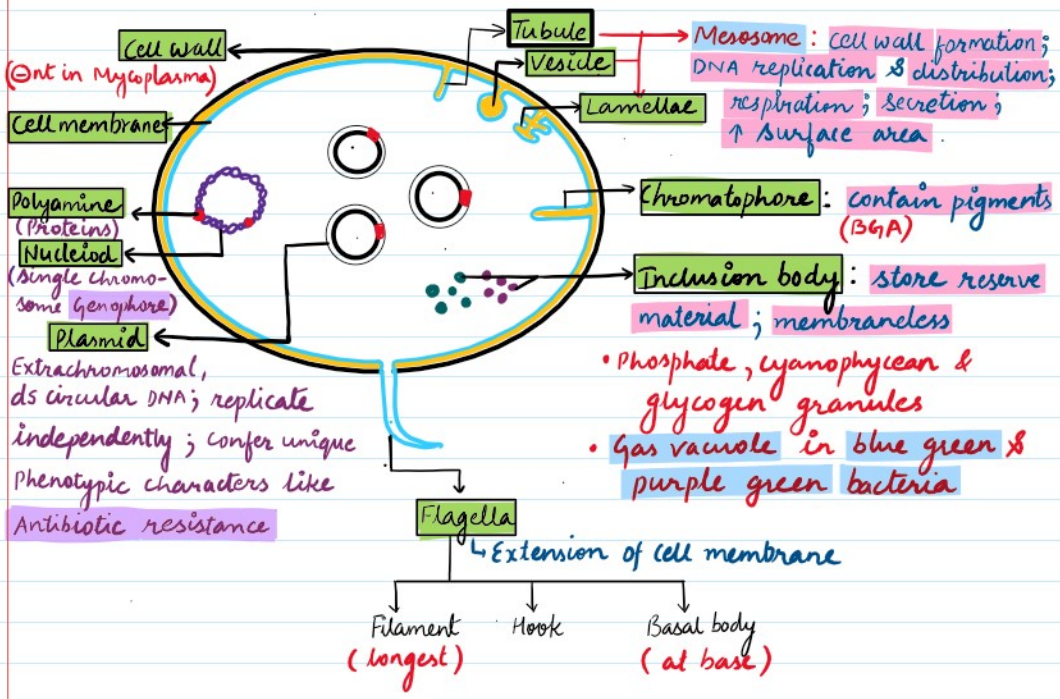


PROKARYOTIC CELL

→ Represented by **Bacteria**, **Cyanobacteria**, **Mycoplasma**, **PPLO**
→ Smaller & divide faster than eukaryotic cells.
→ They may vary greatly in shape and size.

4 basic shapes of bacteria

- **Cocci** (spherical)
- **Bacillus** (Rod shaped)
- **Spirillum** (Spiral)
- **Vibrio** (Comma)



Other surface structures in prokaryotes

- **Pili**: Tubular structure made of pilin **protein**. helps in conjugation (transfer of plasmid from one bacterium to the other)
- **Fimbrae**: Bristle like; give grip to bacteria; help in attaching the bacteria to rock & also to the host tissue.

CELL ENVELOPE

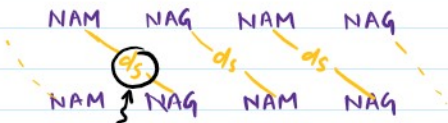
- Present in most prokaryotes, particularly bacterial cells
- chemically complex: **overall protective**
- consist of: ↓

• consist of :

Glycocalyx
- outermost
- defensive
- Polysaccharide macromolecules

- **Capsule**
  - Thick & tough
  - Give protection
- **Slime layer**
  - Loose sheath
  - Protect cell from loss of water & nutrient

Cell wall
- Determine shape of cell
- Give strong structural support
- Prevent bacterium from brusting and collapsing.
- Composition : Murien (peptidoglycan / mucopeptide)



Penicillin inhibits disulphide bond
cause lysis of peptidoglycan

Cell membrane
- Selectively permeable
- similar to eukaryotes
- Internal invagination
  - Mesosomes, chromatophore

Bacteria

On the basis of nature of cell envelope

Gram ⊕ve

Ex. Streptococcus
Staphylococcus
Bacillus

- Thick cell
- Retain crystal violet stain

Gram ⊖ve

- E. coli,
  Pseudomonas
  Helicobacterium pylori

- Thin CW, more porus
- Incapable in retaining CV stain.

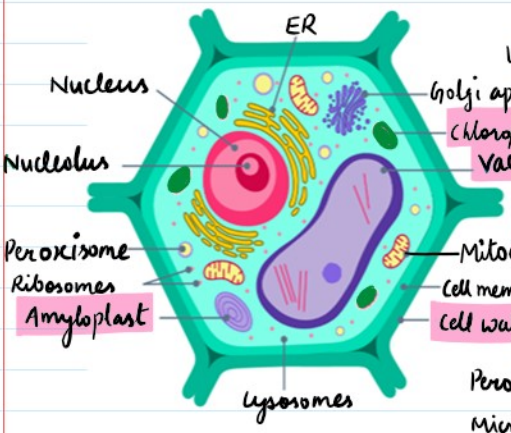
EUKARYOTIC CELL

- Include all the protista, fungi, plants and animals.
- Have extensive compartmentalization & have complex locomotary & cytoskeletal elements.
  - locomotary → Tubulin
  - cytoskeletal elements → Network of filamentous proteins; provide shape; support; motility.

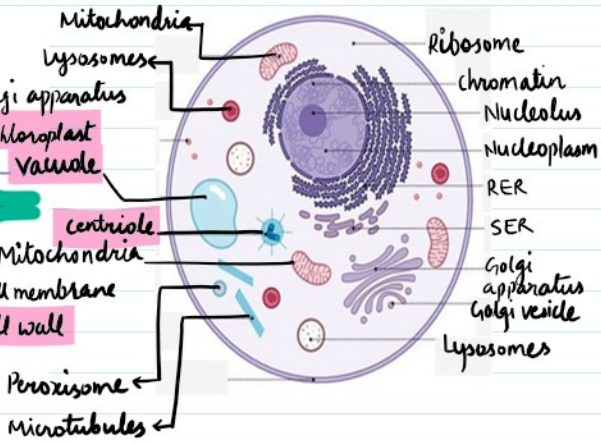
Network of filamentous proteins ; provide shape; support ; motility.

- Genetic material is organised into chromosomes.
- All eukaryotic cells are not identical.

Plant cells



Animal cells



- Larger
- Cell wall ⊕nt
- Vacuole – Large, single, (mostly) central, permanent
- Nucleus – peripheral
- Plastids ⊕nt
- Plasmodesmata ⊕nt
- Centriole ⊖nt

- Smaller
- C.W ⊖nt
- Vacuole → small, many, peripheral, temporary
- Central
- Plastids ⊖nt
- Plasmodesmata ⊖nt
- Centriole ⊕nt

CELL WALL

- Non-living, Rigid structure.
- Outer covering for CM of Fungi and Plants.
- Give shape.
- Protection from mechanical damage and infection.
- It also helps in cell-to-cell interaction.
- Provide barrier to undesirable macromolecules.

Composition of CW

- Fungi: Chitin
- Algae: Cellulose

Fungi
Chitin

Bacteria
Peptidoglycans

Algae
Cellulose, galactans, mannans, $CaCO_3$

Plants
Cellulose, hemicellulose, Pectins, Proteins

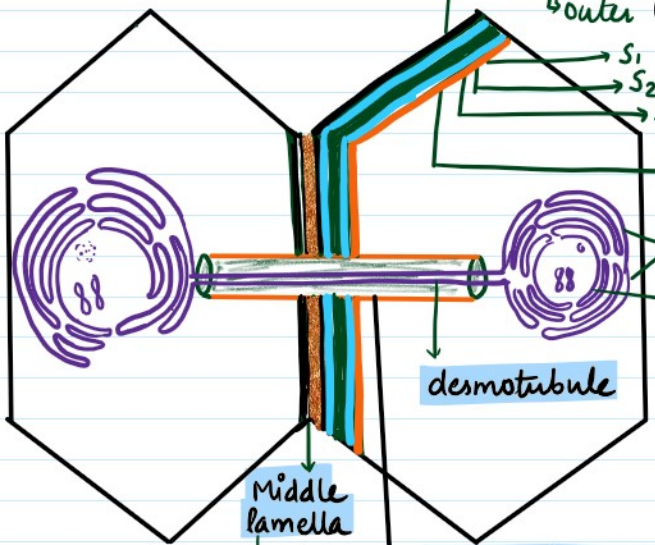
Plant Cell Wall

Middle lamella
- → connects 2 neighbouring plant cells
- → consist of Ca-Mg pectate (Mainly Ca pectate)

Primary wall
- Thin, flexible
- Cellulose, hemicellulose (More), pectin
- ⊕nt in young, growing cells (meristematic & parenchyma cells)

Secondary wall
- Cellulose (more)
- lignified or suberised cell wall
- ⊕ in xylem, fibres, Sclerenchyma

Tertiary wall
- Xylan
- rigid & hard
- Tracheids of gymnosperms.



Middle lamella
- outer layer of tissue
- Formed during telophase

Plasmodesmata
- Symplastic connections b/w 2 plant cells
- ⊕nt in 1°CW

- Formed during telophase
- b/w 2 plant cells
- ⊕nt in 1°CW → living structure

Mature fruit → Middle lamella dissolve → Ripening

Pectinase enzyme

Activated by ethylene & ethephon

Regions where 2°CW ⊖nt, pits are ⊕nt

Simple
- Pits having cavity of uniform diameter

Bordered
- Flask-shaped pits (Tracheids)

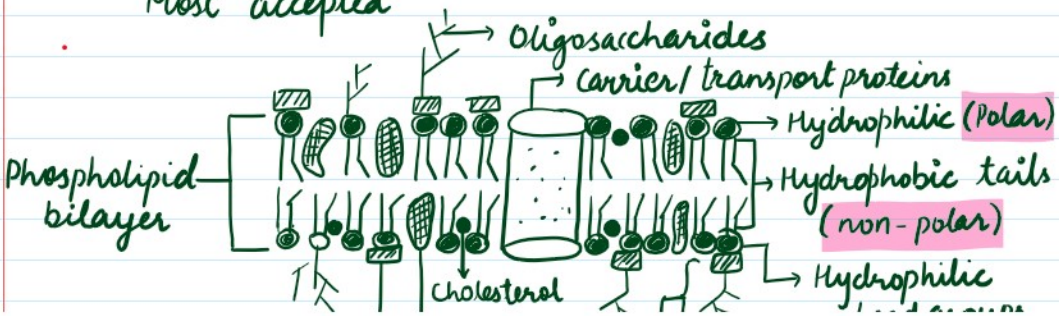
PLASMA MEMBRANE

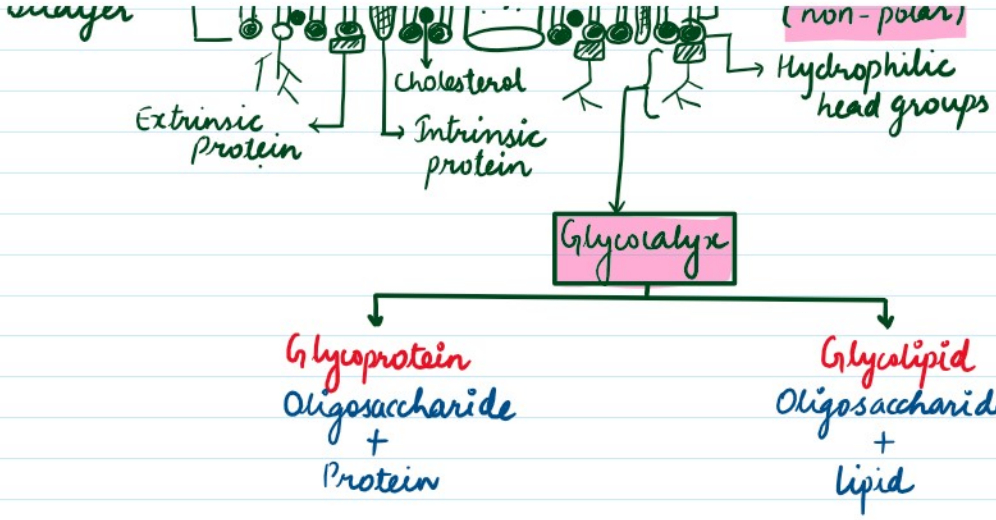
OR Cell membrane ; Biomembrane ; Plasmalemma ; Pellicle ; Unit membrane

- Important for all living organisms.
- Maintain relation of cell with external environment.
- PM of human RBC is used as specimen model for chemical studies of PM.
- The detailed structure of PM studied only after the advent of electron microscope in 1950s.
- Different models given overtime :→

1. Unilayer model – Overton
2. Bilayer model – Gortel & Grandel
3. Sandwitch model – Deneili & Dowson
4. Fluid Mosaic Model – Singer & Nicolson (1972)

→ Most accepted





Glycocalyx

- Glycoprotein: Oligosaccharide + Protein
- Glycolipid: Oligosaccharide + Lipid

Composition Of P.M

RBC

- **Protein** ≈ 52 %
  - Elasticity, Mosaicness
  - Enzymatic Proteins
  - Carrier Protein
  - Glycoprotein ⊕nt on outer membrane
  - Structural Protein
    - **Extrinsic** 30 %
      - RBC membrane - Spectrin
      - Inner memb. of mitochondria - ATPase
      - Easy to remove
      - Give strength to PM → Provide endoskeletal
    - **Intrinsic** 70 %
      - RBC - Rhodopsin
      - Mitochondria inner membrane: cytochrome c oxidase
      - not easy to remove
- **Lipid** 40 %
- **Carbohydrates** <1%

NOTE :- Inner membrane (towards cytoplasm) is rich in extrinsic proteins.

LIPIDS

- Phospholipids
  - Polar head (hydrophilic)
  - Non-polar
- Glycolipids
- Sterols
  - Cholesterol
    - Animals
  - Phytosterol
    - Plants
  - Ergosterol
    - Protista

(hydrophilic)

Non-polar tail (hydrophobic)

Cholesterol
- Animals

Phytosterol
- Plants

Ergosterol
- Protista, Fungi

Movements of Phospholipids

Flexing movement in fatty tails



Lateral Exchange b/w nearby phospholipids of same layer



Flip-flop Exchange b/w phospholipids of diff. layers



Rotational movement of phospholip-ids on their own axis



TRANSPORT ACROSS PLASMA MEMBRANE

Active
Against concentration gradient
Low → High
Ex. $Na^{2+} - K^{+}$ pump

Passive
Along conc^{n} gradient
High → Low
Osmosis, Diffusion

Bulk
For transport of macromole-cules

- **Exocytosis** out-of cell
- **Endocytosis** Into cell
  - **Pinocytosis** liquid intake
  - **Phagocytosis** solid intake

Special Modification of Cell Membrane

- **In-Folding**
  - Microvilli
  - Lamellipodia
- **Out-folding**
  - Stereocilia
  - ...

Mesosome
Bacteria

Lamellosome
Insect cells

Lamasome
Fungi

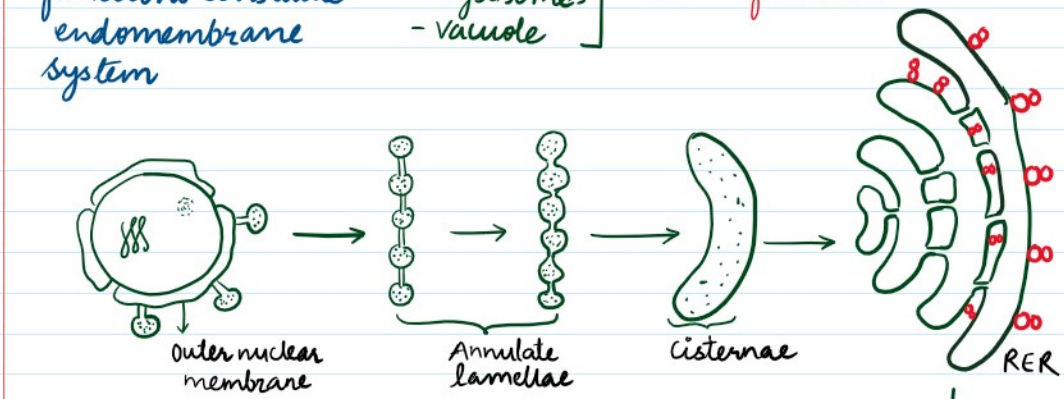
Stereocilia
sensory organ

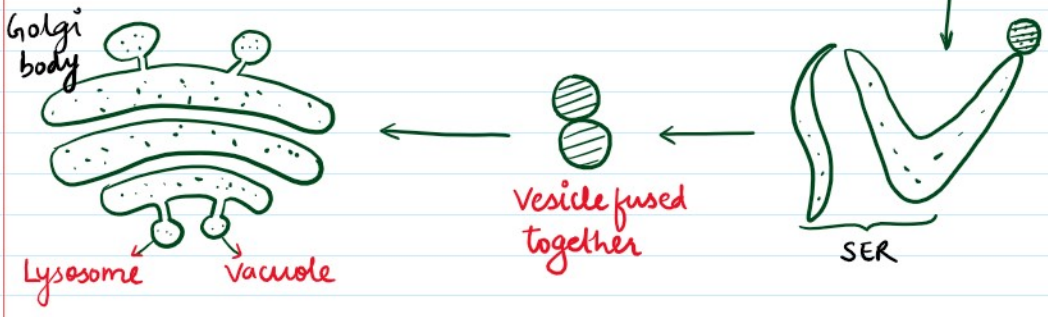
Microvilli
Intestine

Endomembrane System

- Membranous organelles which are coordinated in functions constitute endomembrane system
  - ER
  - Golgi
  - lysosomes
  - Vacuole

} different in structure & functions





ENDOPLASMIC RETICULUM

Discovered by Porter.
⊕nt in eukaryotes.
Less developed in Meristem.
Network of scattered tiny tubular structures.
ER divides the intracellular space into 2 distinct compartments
- luminal (inside ER)
- Extra-luminal (cytoplasm)



- Ribosomes ⊕nt
- Rough

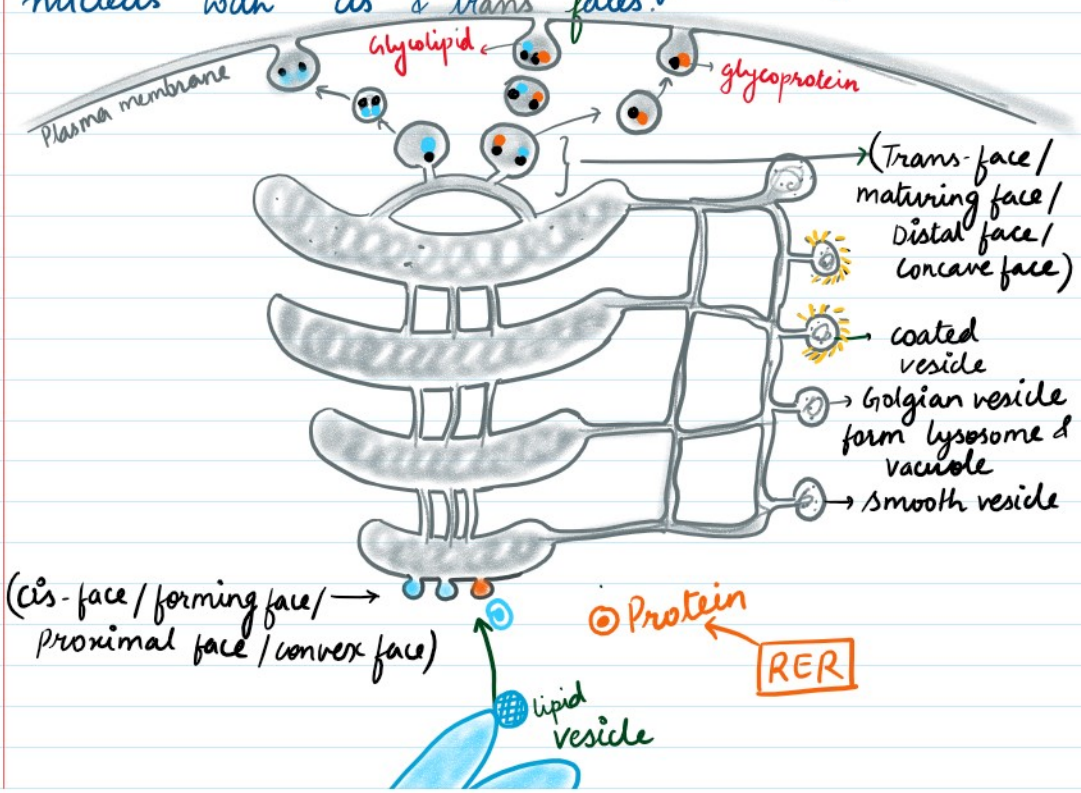
SER ←

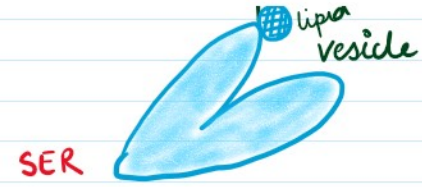
- Ribosome ⊖ nt
- Smooth
- In cytoplasm, away from nucleus.
- Major site for lipid synthesis.
  In animal cells]
  lipid-like steriodal hormone synthesis
- Detoxification of toxic substances by Cytochrome P_{450}.
- Liver cells, adipose cells, adrenal cells of kidney are rich in SER.

- Ribosomes ⊕ nt
- Rough
- Extensive & continuous with outer membrane of nucleus.
- Frequently observed in cells actively involved in protein synthesis & secretion.

GOLGI APPARATUS

- They are also known as dictyosomes (In Plants – cisternae are not connected)
- Discovered by Camillo Golgi.
- ⊕nt in all Eukaryotes (Except: Mature RBC, sieve tubes).
- Well developed in animal cells.
- Less developed in plant cells.
- They consist of many flat, disc-shaped sacs or cisternae of 0.5 µm to 1.0 µm diameter.
- The Golgi cisternae are concentrically arranged near nucleus with cis & trans faces.





- **Structure** of Golgi complex – Cisternae
  – tubules
  – vesicles
- **Functions** :- Modification
  Packaging
  Secretion / transportation } of materials → lipids, Proteins, Ribosomes
  - Important site for formation of glycoproteins & glycolipids
  - Acrosome of sperm is rich in golgi bodies.
  - Formation of PM during cell division.

LYSOSOMES

- Membrane bound vesicular structure.
- Formed by the process of packaging in the golgi apparatus.
- Isolated lysosomal vesicles are rich in almost all type of hydrolytic enzymes.
  (lipases, proteases, carbohydrases)
  → optimally active at acidic pH.
  → enzymes are capable in digesting lipids, proteins, carbohydrates and nucleic acids.
- They show **polymorphism**.

Primary lysosomes	Secondary lysosomes	Residual bodies	Autophagic vacuole
• Newly formed small vesicles ⊕nt on trans-face of golgi • Inactive enzymes ⊕nt	OR • Heterolysome / digestive vesicles • (Phagosomes + 1° lysosome) → rich in digestive enzymes	• 2° lysosomes in which digestive enzymes ⊕nt • Residual vesicles + PM → Perform Euphagy	• Suicidal bags / autolytic • Many 1° lysosomes surrounding dead / old cells. Autolysis of larval organs during development

during development

VACUOLES

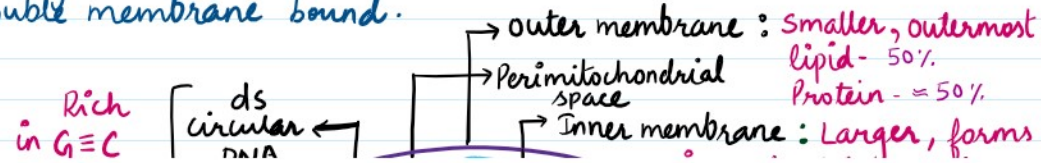
- Membrane bound space found in the cytoplasm.
- contains – water
  - sap
  - excretory products
  - other materials not useful for cells
- Bound by single membrane (Tonoplast)
- In plant cells – vacuole can occupy upto 90% of volume of the cell.
  - * – Tonoplast facilitates the transport of a number of ions & other materials against concentration gradients into the vacuole.
    conc^n is higher in vacuole than cytoplasm

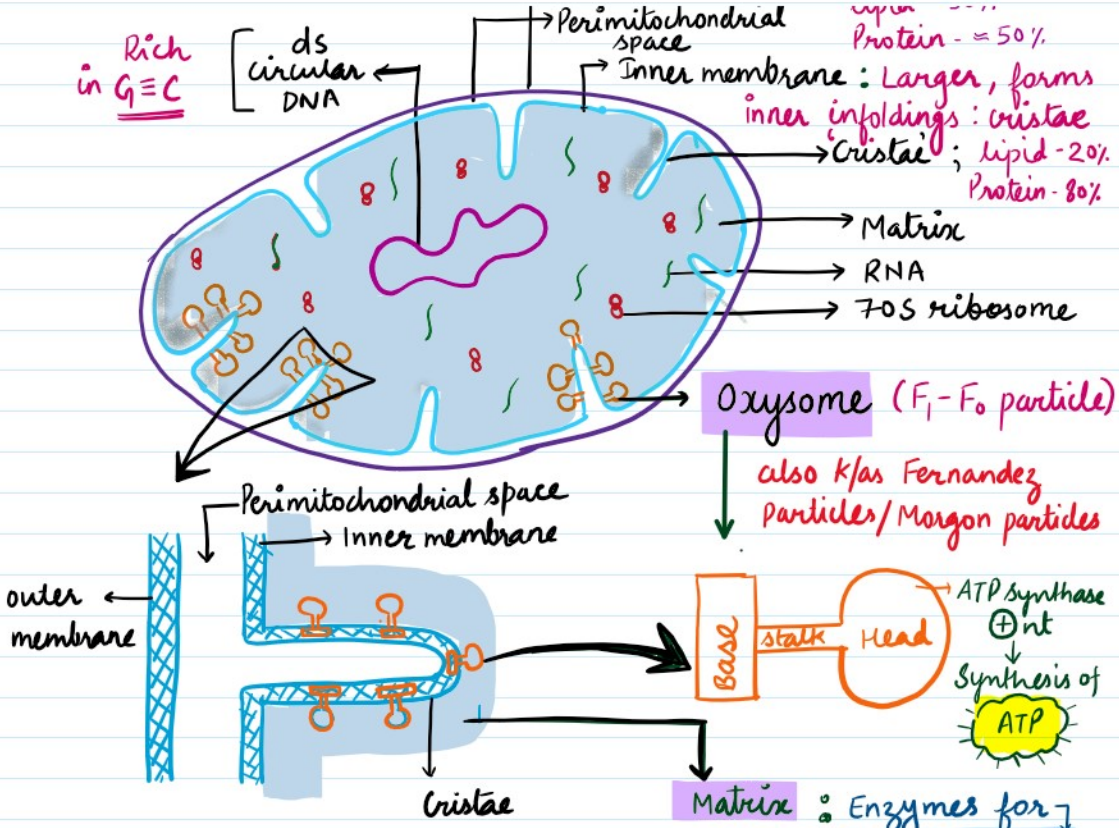
Types of Vacuole

Sap vacuole	Food vacuole	Contractile vacuole	Gas vacuole
- Plants - Maintain turgidity	Protists - Engulfing food particles	Amoeba Osmoregulation Excretion	Bacteria Reserve material

MITOCHONDRIA

- Discovered by Kolliker in insects muscle cells.
- Term given by Benda.
- Altman termed it bioplast.
- Janus green vital stain is used to locate mitochondria.
- No. of mitochondria per cell varies with the physiological activity of cell.
- Typically it is sausage-shaped or cylindrical.
- diameter 0.2 – 1.0 μm (average 0.5 μm)
  length 1.0 – 4.1 μm
- Double membrane bound.





Protein - ≈ 50%

: Larger, forms inner infoldings : cristae ; lipid - 20%, Protein - 80%

- Site for aerobic respiration – Kreb's cycle
- Site for ATP synthesis – 'Powerhouse of cell'.
- Mitochondria divide by Fission.
- e^- carrier & enzymes for ATP synthesis found on inner mitochondrial membrane.
- Mitochondria without outer membrane is k/as Mitoplast.
- Mitochondria is semi-autonomous organelle

Able to synthesize some of its own proteins! ←
- ds circular DNA
- RNA
- Ribosomes (70S)

PLASTIDS

→ Easily visible under microscope
→ ⊕nt in all plants & euglenoids

Coloured

Choloplasts
green-coloured
- Chlorophyll ⊕nt
- Photosynthesis

Chromoplasts
Yellow, red, orange...
- Carotenoids ⊕nt
- Pollination - to

Colourless (Leucoplasts)
- Granum ⊖nt
- Found near nucleus
- For storage

→ Amyloplast: Starch
Ex. Potato tuber, Rice

green-coloured
- Chlorophyll ⊕nt
- Photosynthesis

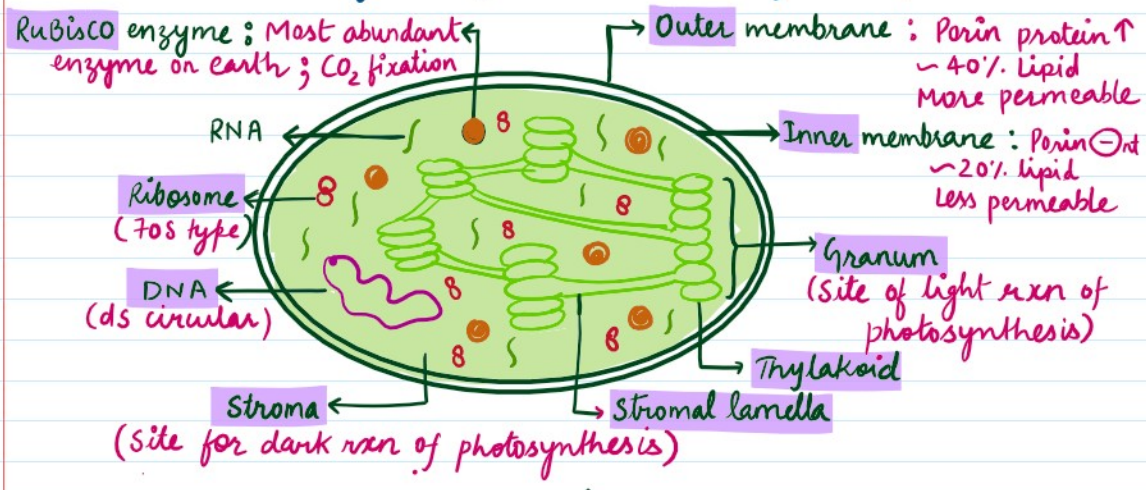
yellow, red, orange...
- Carotenoids ⊕nt
- Pollination - to attract pollinators
- Ripened fruits, vegetables, petals

- for storage
→ Amyloplast: Starch
Ex. Potato tuber, Rice

→ Aleuroplast: Protein
Ex. Aleurone layer of Maize

→ Elaioplast: Lipid / Oil
Ex. Castor seed

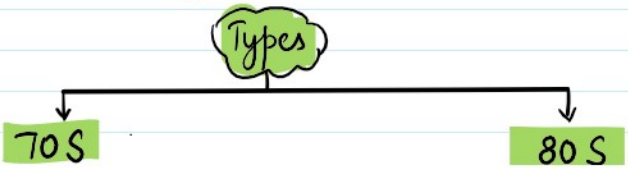
- Green parts of plants
↓
Mesophyll cells (majority)
- These are lens-shaped, oval, spherical, discoid or even ribbon-like.
- Variable length (5-10 μm)
width (2-4 μm)
- Their no. varies from (1/cell - Chlamydomonas), a green alga to (20-40/cell - mesophylls).
- Double-membrane bound organelle.
- Inner membrane of chloroplast is relatively less permeable.

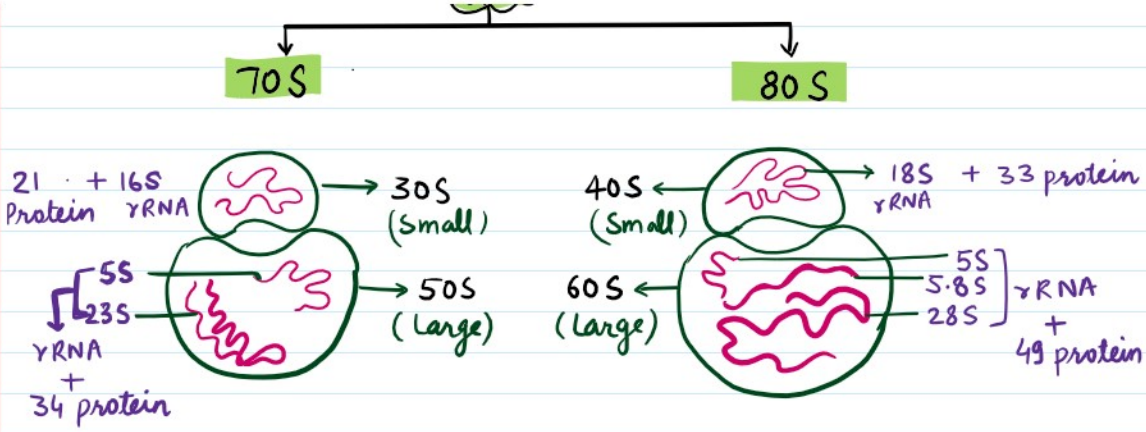


NOTE: Mitochondria & chloroplasts are self-duplicating organelles.
Formed by fission.
Both are semi-autonomous cell organelles, as they have - RNA, circular DNA, 70S ribosomes.

RIBOSOMES

- Granular structures
- First observed under the e⁻ microscope as dense particles by George Palade (1953).
- Composed of: RNA + Proteins.
- membrane-less organelle.





- Found in prokaryotes, mitochondria chloroplast
- Found in cytoplasm of eukaryotes.

- Function : Site for protein synthesis hence K/as Protein factory.

- Both (large & small) units of ribosomes joined together by specific conc^{n} (0.001 M) of Mg^{2+} ions.
- 'S' stands for (Sedimentation coefficient / Svedberg unit)
  ↳ Indirect major of density & size)

- Ribosomes connected together to form dimer.
- Many ribosomes aligned in series on single mRNA to form polysome / polyribosome during protein synthesis.
- Smallest cell organelle within cell.
- ⊕nt in largest number in the cell.

CYTOSKELETON

- Form elaborate network of filamentous proteinaceous structures.
- Found only in eukaryotic cells.
- Provide mechanical support, motility, maintenance of cell shape
- Consist of

Microtubules

⊕ in cytoplasmic matrix of eukaryotes, cilia, flagella, centriole
- decide plane of cw

Microfilaments

- ⊕ in eukaryotic cells
- Solid, unbranched, filamentous.
- Consist of actin (mainly) & myosin

Intermediate filaments

- They are non-contractile filaments of basic or acidic proteins

centriole
- decide plane of CW formation.
- Frame str. of cell
- Maintain shape & size
- Hollow, unbranched, cylindrical
- Tubulin protein

- consist of actin (mainly) & myosin
- Provide support to PM.
- They represents contractile network of cell.
- Pseudopodia formation
- Microvilli
- Cleavage furrow formation during cell division

filaments of basic or acidic proteins
- Form hair & nails
- support nuclear envelope and help to organise chromatin.

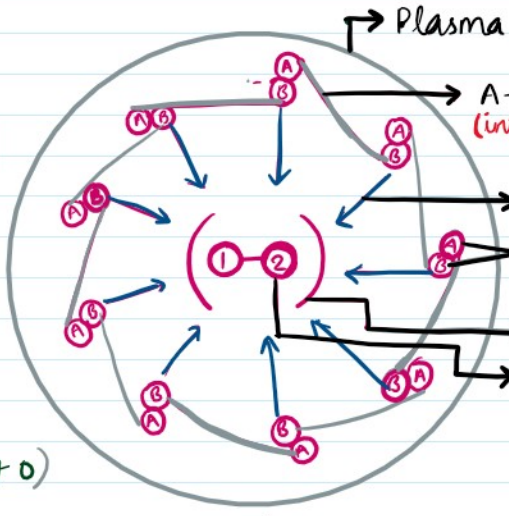
CILIA & FLAGELLA

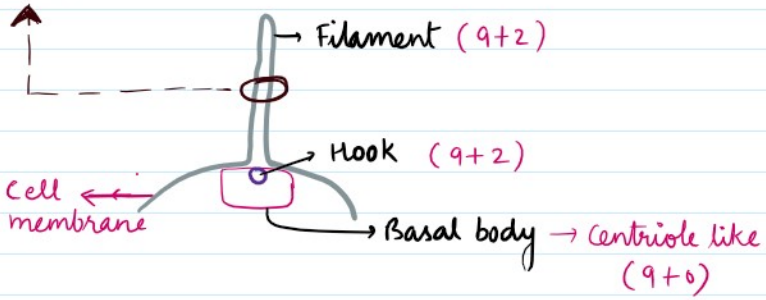
- Short, hair-like structure.
- only found in eukaryotic cells.
- occur throughout the cell (prokaryotes)
- Beat in coordination (oar-like)
- Indicate sweeping movement.
- Involve in processes like feeding, locomotion & circulation
- found in lining of body tubes.

- long, whip-like structures.
- found in eukaryotic cells. (Prokaryotic flagella diff. from eukaryotic)
- Occur at one end of a cell.
- Beat independently
- Indicate undulatory movement.
- Involve in locomotion
- found in most bacteria, archaea, eukaryotes.

- core of cilia & flagella – Axoneme
- The arrangement of microtubules in cilia & flagella is (9+2)

Except: prokaryotic flagella (9+0)



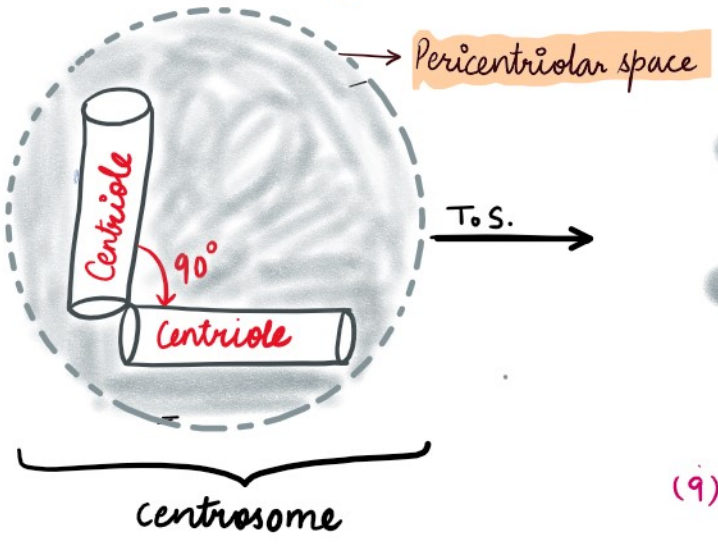


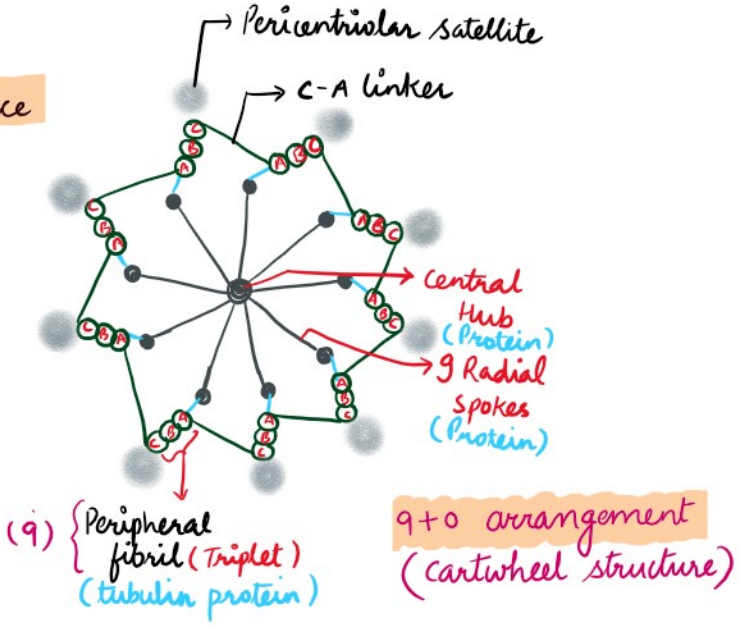
CENTROSOME and CENTRIOLES

- Non-membranous cell organelle
- Present in eukaryotes except: plant cells

- Surrounded by amorphous pericentriolar materials.
- They are microtubule generator molecules.
- Responsible for assembly and dissembly of microtubules during spindle formation.

Function :- Help in cilia & flagella formation (basal body)
- spindle formation
- astral rays formation

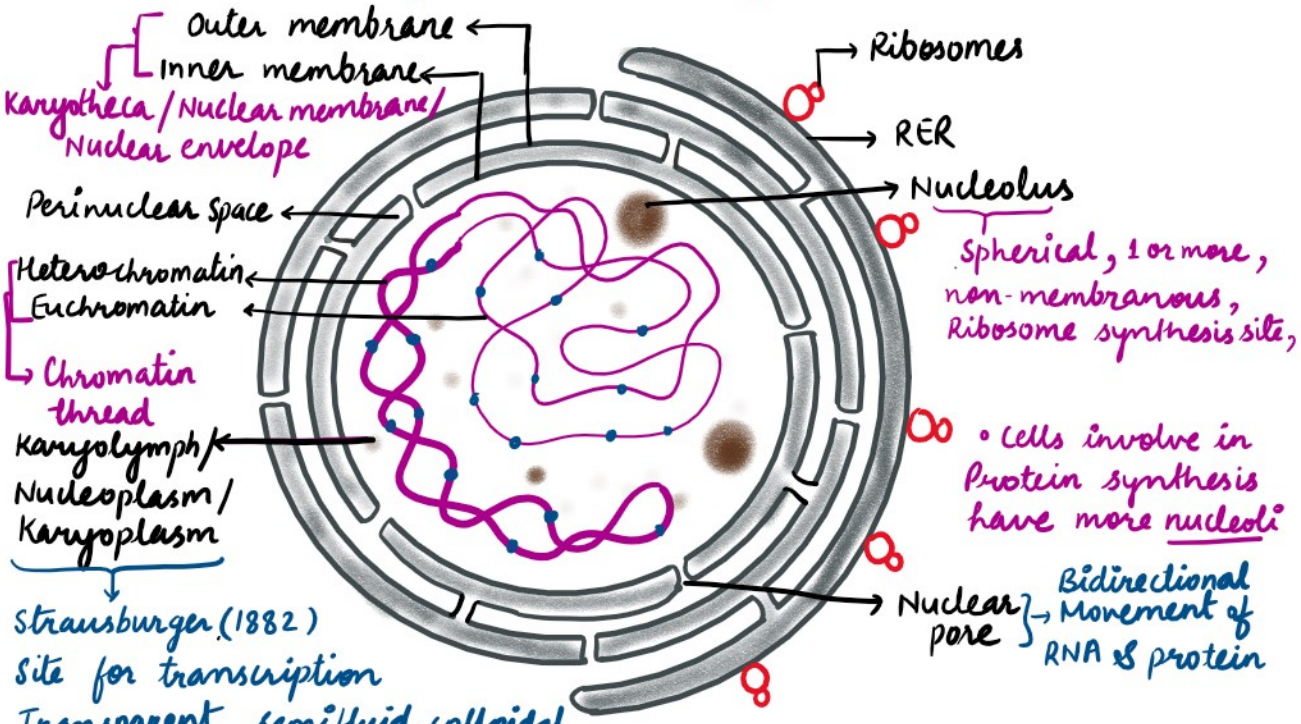




NUCLEUS

- Nucleus as a cell organelle was first described by Robert Brown (1831)
- The material of the nucleus stained by basic dyes was given the name chromatin by Flemming.
- Larger than other organelles
- Control almost all the cellular functions (Brain of cell).

- Larger than other "organelles"
- Control almost all the cellular functions (Brain of cell).
- Some cells have more than 1 nucleus.
  Multinucleate – Opalina
  Binucleate – Paramoecium
  Anucleate – RBC, sieve tubes Except: Camel & Lama

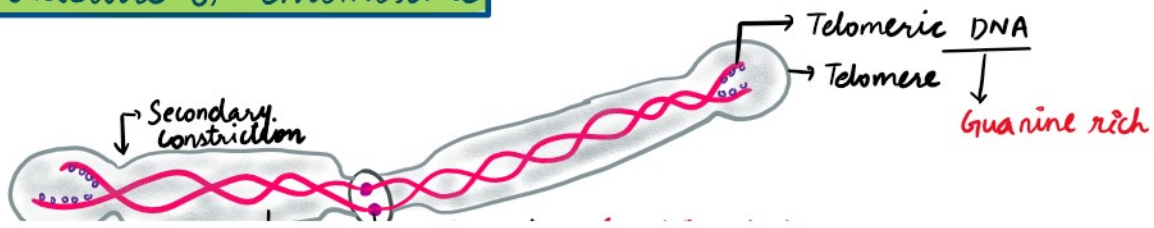


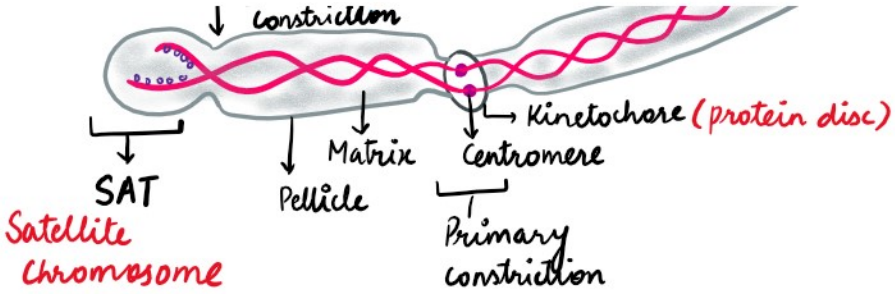
- Strausburger (1882)
- Site for transcription
- Transparent, semifluid, colloidal
- pH = 7.4 ± 0.2

- **Chromatin** :· (Chroma = colour) → Acetocarmine
  - The interphase nucleus has a loose and indistinct network of nucleoprotein fibres called chromatin.
  - Contain (DNA = ≃ 33%), (Histones = ≃ 35%), (Non-histone = ≃ 20%), (RNA = ≃ 5%.)
  - Interphase (chromatin) ⟶ Karyokinesis (chromosomes)
  - A single human cell has approx. 2 metre long thread

46 Chromosomes

Structure of Chromosome

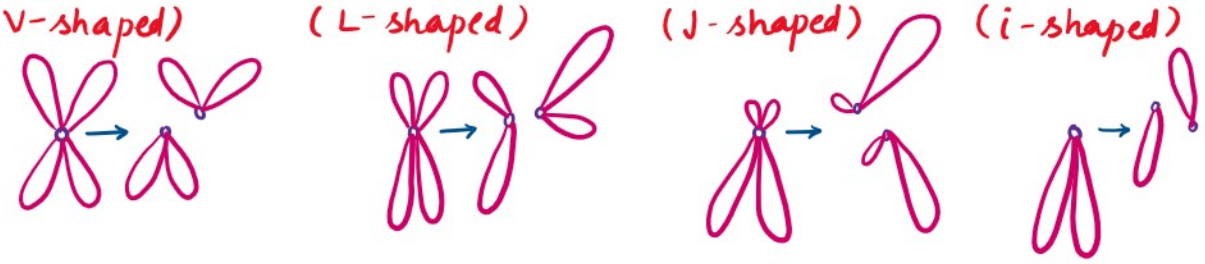




Types of Chromosomes

On the basis of position of centromere visible during Anaphase

- **Metacentric** (V-shaped)
- **Sub-metacentric** (L-shaped)
- **Acrocentric** (J-shaped)
- **Telocentric** (i-shaped)



NOTE → In humans, telocentric chromosomes are ⊖nt.

Microbodies

- Membrane-bound; minute vesicles
- Contain various enzymes
- ⊕nt in both plant and animal cells

Peroxisome

- Rich in catalase enzyme

Spherosome

- From SER
- Plant lysosome
- Hydrolytic enzymes
- Involve in synthesis & storage of lipids

Glyoxysomes

- Enzyme for Glyoxylate cycle (Gluconeogenesis)

Fat → Carbohydrate

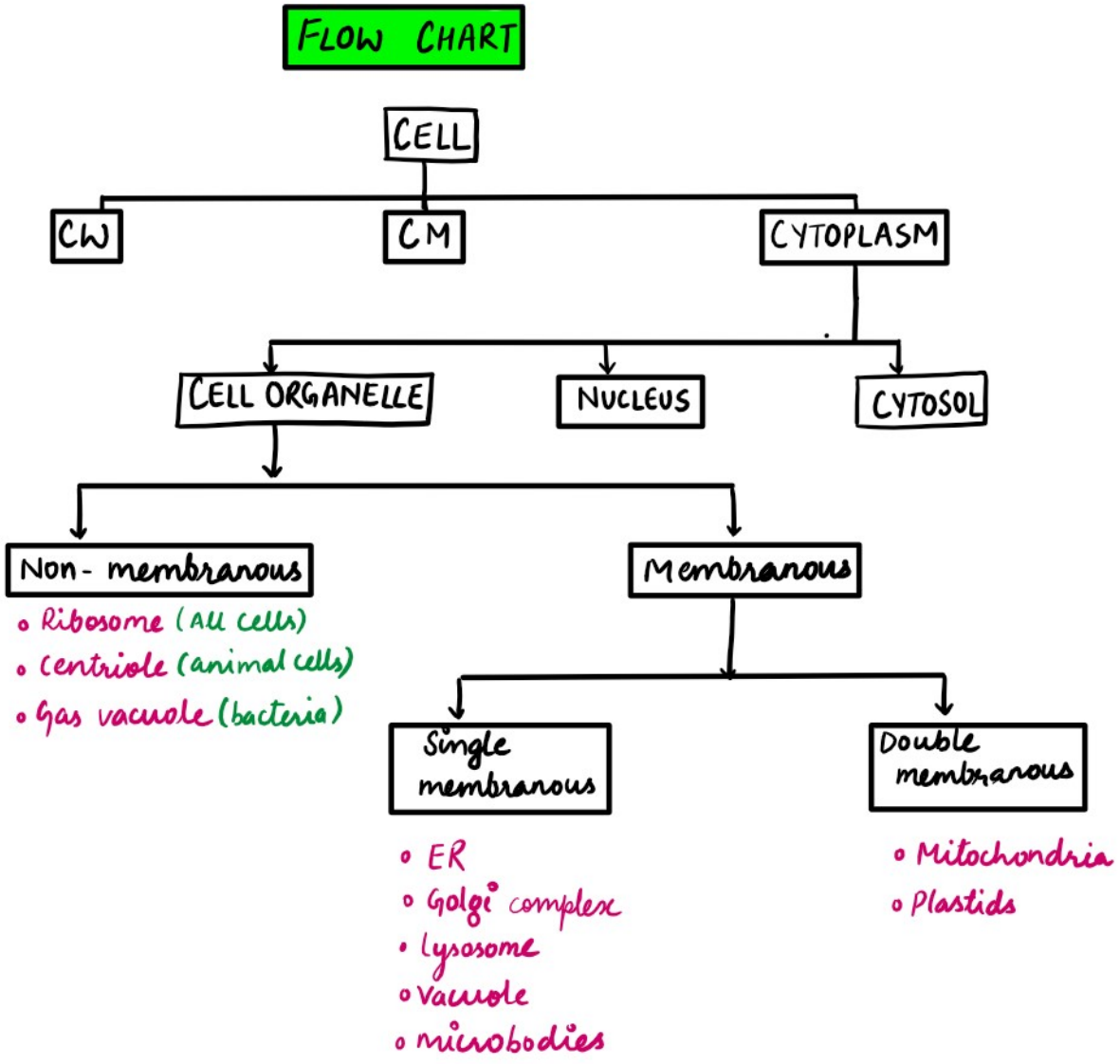
- Germinating fatty seeds.

enzyme

- Hydrolytic enzymes
- Involve in synthesis & storage of lipids

cycle (gluconeogenesis)
Fat → Carbohydrate

- Germinating fatty seeds.



- Ribosome (All cells)
- Centriole (animal cells)
- Gas vacuole (bacteria)

- ER
- Golgi complex
- Lysosome
- Vacuole
- Microbodies

- Mitochondria
- Plastids